

Loyola Marymount University

CMSI 3300 Artificial Intelligence

Fall 2024

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Announcements

- Homework 1
  - Due Friday 10/4 @5 pm
  - Progress:
    - Finish MazeClause (Part 1.1)
    - Start on MazeKB (Part 1.2)

This Lecture

- Bayesian Networks
  - Structures
  - Semantics
  - Markovian Factorization
- Homework 1

Review

Propositional Logic

Limitation 1: **Uncertainty**

Probabilistic Logic

Conditioning Rule $P(Q|e) = \frac{P(Q,e)}{P(e)}$

Truth Table

World	L	T
w_0	F	F
w_1	F	T
w_2	T	F
w_3	T	T

Unscalable

Joint Probability $P(Q, e_1, e_2, \dots)$

Symbolic Inference

Limitation 2: **Scalability**

Bayesian Networks

Joint Independence **Factoring**

$A \perp\!\!\!\perp B \iff P(A,B) = P(A)P(B)$

Theorem Proving

1. $R \Rightarrow W$
2. R

3. W

Last Lecture - Bayesian Networks (BN)

- Two Components
  - Structure
    - DAG: Directed Acyclic Graph
    - Encode **independence** (e.g., intuitive [cause] -> [effect] or [parent]->[child]) relationships between variables/nodes
  - Semantics
    - Factor **joint** probability (JPT) using **conditional** probability (CPT) and independence relationships

Not necessarily to be causal in BN

Structure — Model Sources

- Where do Bayesian Networks come from?
  1. **Top-down** [Model -> Data]
     - Start with a programmer-curated **model** and then **interpret** witnessed **data** through the model.
  2. **Bottom-up** [Data -> Model]:
     - Start with **data** and attempt to **learn** the **model** (i.e., structure relationships between variables) based on witnessed independence relationships.

Semantics

- Conditioning Rule:
  - $P(Q|e) = \frac{P(Q,e)}{P(e)}$ joint -> conditional



  - $P(Q,e) = P(Q|e)P(e)$ conditional -> joint

{% of Q and e} = {% of Q in worlds where e is true}{% of a world where e true}

Independence and Conditioning

- $Q \perp\!\!\!\perp e \Leftrightarrow P(Q \mid e) = P(Q)$
  - Example: two fair dice A and B
    - $A \perp\!\!\!\perp B \Leftrightarrow P(A|B) = P(A)$

"dropping e": knowing about "e" tells me nothing about "Q"

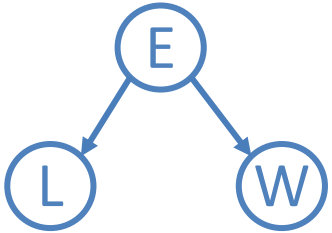
- $Q \perp\!\!\!\perp e_1 \mid e_2 \Leftrightarrow \quad P(Q \mid e_1, e_2) = P(Q|e_2)$

Conditional Independence

"dropping e_1": knowing about "e_1" tells me nothing about "Q" that "e_2" did not already tell me

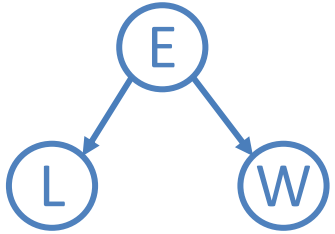
Independence as Flow of Information

- Earthquake
  - E: an earthquake was detected
  - L: call from LMU
  - W: wireless emergency alert



- $E \not\perp\!\!\!\perp L$ and $E \not\perp\!\!\!\perp W$. Why?
  - There is flow of information from E->L and E-> W
- $L \perp\!\!\!\perp W \mid E$
  - There is **no** flow of information from L-> W or W-> L. Why?
  - Knowing about W tells me nothing about L that "E" did not already tell me

Why conditional independence?



- Example:
  - 3 variables L, W, E and $L \perp\!\!\!\perp W \mid E$
  - How to factor $P(L, W, E)$ using **conditioning rule**?

Conditioning Rule:
$P(A, B) = P(A|B)P(B)$

$$P(L, W, E) = P(L|W, E)P(W, E)$$
$$= P(L|W, E)P(W|E)P(E)$$

  - How to reduce it using **conditional independence**?

$$P(L, W, E) = P(L|W, E)P(W|E)P(E)$$
$$= P(L|E)P(W|E)P(E)$$

Conditional Indep.
$P(A|B, C) = P(A|C)$

$P(child|parent) = P(effect|cause)$

Markovian Factorization

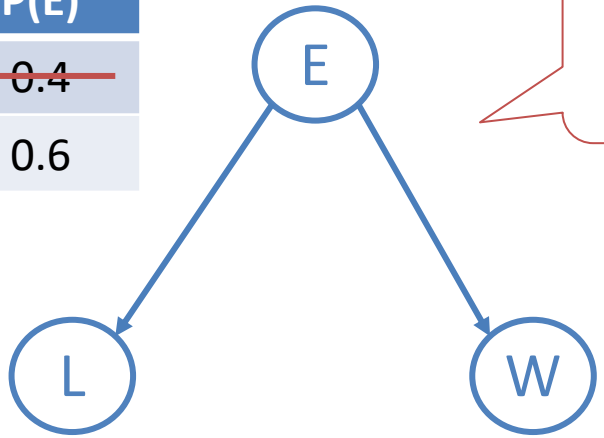
- **Goal:** every node in BN only needs to know its **CPT** (i.e., P(child | parent) or P(effect | cause)), and the **product of CPTs** allows us to **reconstruct** their JPT
- Formally, the Markovian Factorization of the joint distribution is:

$$P(V_1, V_2, \dots) = \prod_{V_i \in \boldsymbol{V}} P(V_i | parents(V_i))$$

{Product of all CPTs in BN}

Example

E	P(E)
~~0~~	~~0.4~~
1	0.6



Given that L, W, E are binary variable, can these tables be reduced?

E	L	P(L|E)
~~0~~	~~0~~	~~0.75~~
0	1	0.25
~~1~~	~~0~~	~~0.17~~
1	1	0.83

E	W	P(W|E)
~~0~~	~~0~~	~~0.9~~
0	1	0.1
~~1~~	~~0~~	~~0.2~~
1	1	0.8

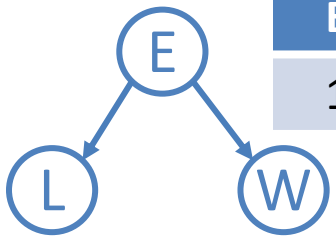
Example

- What is P(E=0)?
  - P(E=0)=1-P(E=1) = 1-0.6 = 0.4
- What is P(E=0, L=1, W=0)?

  = P(L=1|E=0)P(W=0|E=0)P(E=0)

  = 0.25*(1-0.1)*0.4

  =0.09



E	P(E)
1	0.6

E	L	P(L|E)
0	1	0.25
1	1	0.83

E	W	P(W|E)
0	1	0.1
1	1	0.8

Bonus



https://youtu.be/GEFxFVESQXc

Remaining of the lecture

HOMEWORK 1